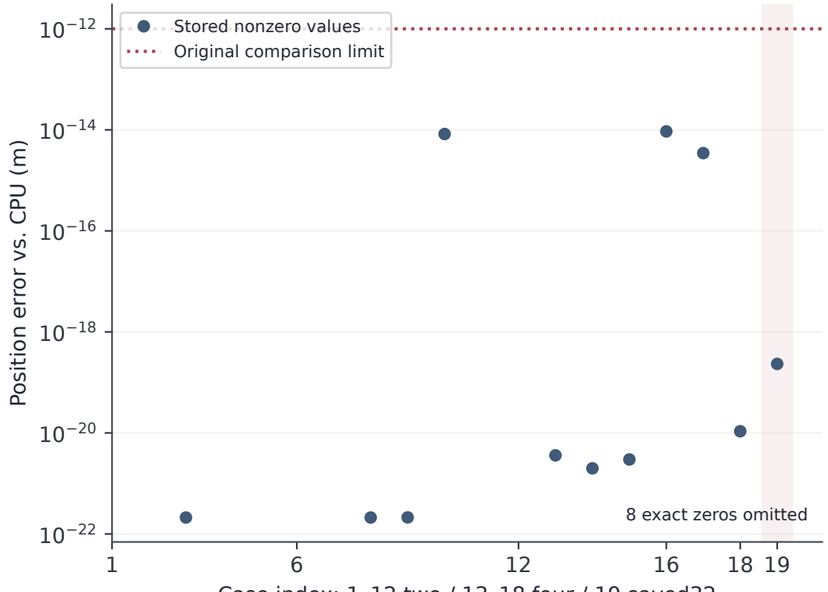
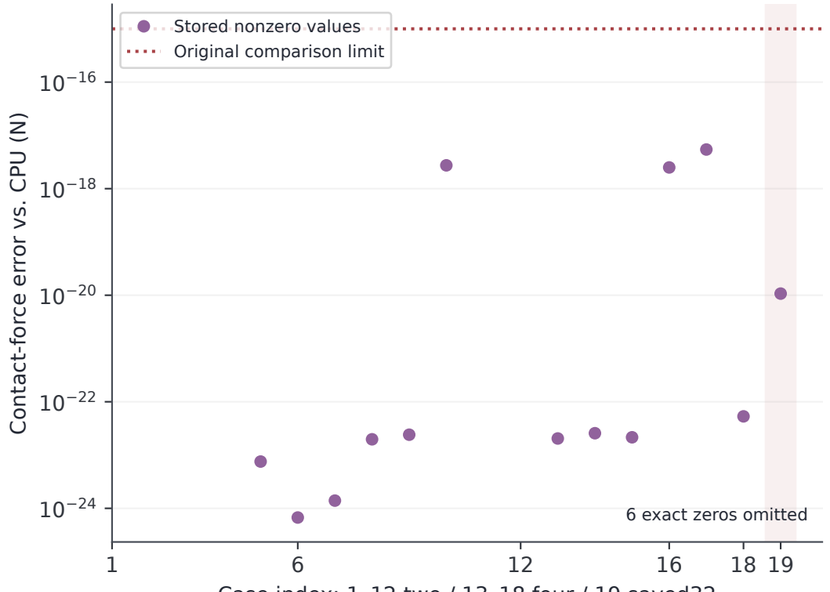
GPU B2 fixed-state prototype: GPU_B2_NOT_ADMITTED

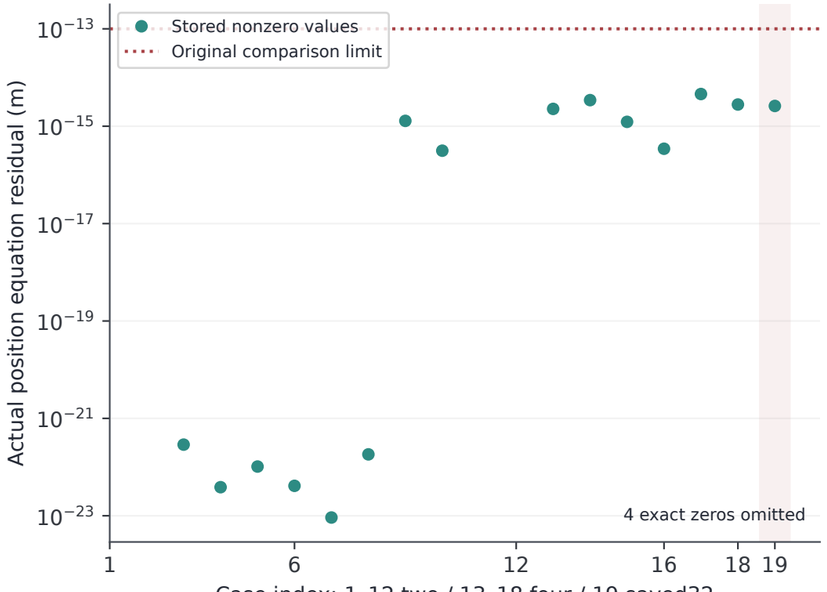
a Position agreement



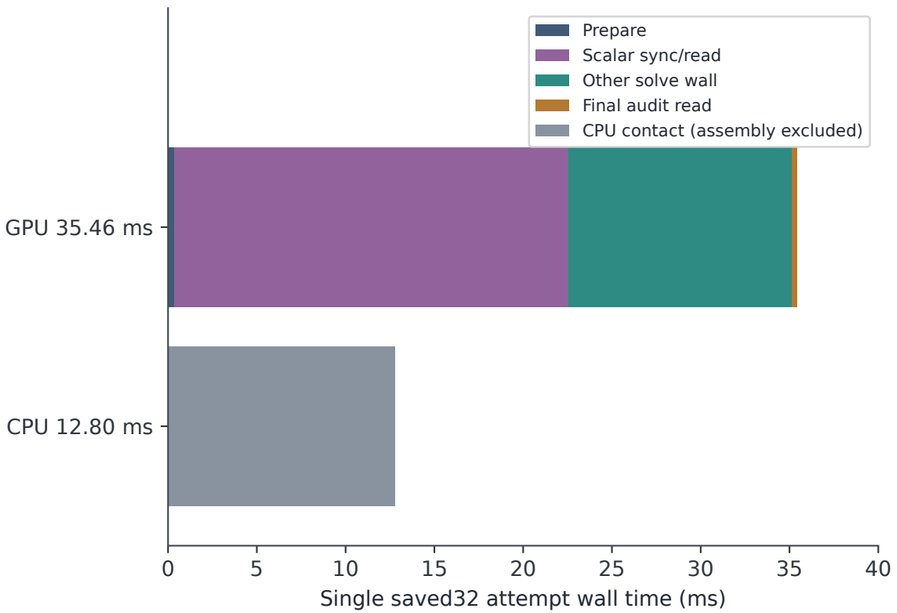
b Force agreement



c Independently checked residual



d Measured attempt costs; not a speed benchmark



e Why the saved32 solver was rejected

Geometry / active contacts / edges: agree
Overlap 17.498 nm < 17.8 nm budget
Correction 1.932 nm < 178 nm budget

Conservative bound: 1.850e−14 m
Registered stopping target: 1e−14 m
Armijo exhausted 20 line-search trials
Base PCG missed its recursive-residual target

18 small cases match the CPU reference;
2 are expected overlap-budget rejections.
No changed tolerance; no production admission.

f Full cost and scope retained

saved32 allocation: 0.534 ms
Reusable PCG graph build: 40.560 ms
35 scalar reads: 22.242 ms within solve
Explicit buffers: 78,376 bytes (lower bound)
CG workspace / context / graphs: extra

Five matched timing repeats: NOT_RUN
Stopped after the solver admission failure
GPU v1 engineering failure also archived

B = 1 fixed-state experiment only
No new 10 ms / B64 / training set
PHYSICS_UNCALIBRATED; training_ready=false

Original budgets retained. Small cases 17–18 intentionally verify the same CPU/GPU overlap rejection; they are not positive contact-quality passes.
GPU comparison agreement does not override nonconvergence. CPU assembly is outside the shown CPU timing; graph construction is outside the GPU step.